UNIVERSIDAD DISTRITAL FRANCISCO JOSE DE CALDAS

Systems Analysis & Design — Workshop No. 1

Smart Campus Parking Management System

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1. Introduction

This report details the design and analysis of the **Smart Campus Parking Management System (SCPMS)**, a technological solution aimed at optimizing motorcycle parking usage at the Universidad Distrital Francisco José de Caldas. Currently, the management of this area in Basement 3 of the main campus faces critical challenges due to a lack of organized structure and the absence of real-time information for users.

The core problem lies in an **information asymmetry**: students and faculty arrive at the facility without knowing if they will find an available spot, resulting in congestion, wasted time, and inefficient use of the installed physical capacity. Data obtained through direct observation and surveys confirm that the majority of users perceive the current system as inefficient and overwhelmingly support the implementation of a digital alternative.

This project proposes a transition from an improvised occupancy model to a structured reservation system integrated into the university's mobile infrastructure. Throughout this report, we will analyze the system's scope, the evidence collection methodology, and the improvement proposals that will transform on-campus mobility, benefiting both the student community and security personnel.

2. System Overview and Scope Definition

2.1 Comprehensive System Description

The Smart Campus Parking Management System (SCPMS) is a proposed digital solution designed to address the chronic inefficiency in motorcycle parking utilization at the Basement 3 (Sótano 3) facility of Universidad Distrital Francisco José de Caldas. The system targets a well-defined operational problem: the absence of demarcated parking cells causes spatial disorder — a single motorcycle can occupy the physical footprint of three designated spaces — resulting in a persistent gap between actual available capacity and perceived capacity by users.

The proposed solution is built around a digital reservation and monitoring platform integrated with the university's existing mobile application. Its core functionality enables students with registered vehicles to view a numbered, real-time map of Basement 3, reserve a specific space for a defined time slot, and receive or relinquish that reservation based on their academic schedule. Security personnel are supported through automated notifications when a reserved space has been violated, reducing the dependency on on-site judgment and intermittent physical supervision.

Main System Components

- **Reservation and Availability Module:** A numbered digital floor plan of Basement 3 exposed through the university app, showing real-time occupancy status for each demarcated space.

- **Time-Slot Scheduling Engine:** Logic that binds reservations to specific academic time blocks (e.g., 06:00–10:00, 12:00–16:00), preventing indefinite occupancy and enabling priority handoffs.
- **User Account Integration:** Linkage with existing university credentials and registered vehicle plates, ensuring each reservation is uniquely attributed and traceable.
- **Security Alert Subsystem:** Automated reporting to security staff when a reserved space is occupied by an unregistered vehicle, or when a confirmed user fails to vacate within the allotted window.
- **External Overflow Advisory Module:** A complementary feature identifying idle reserved spaces during off-peak academic hours, providing guidance to users of informal external parking areas (sidewalks and adjacent roads) so that temporary overflow can be managed in a structured, priority-based manner.

2.2 Scope Definition

Within Scope

- Digital demarcation and management of all motorcycle parking spaces within Basement 3 of the main campus building.
- Real-time visualization of space availability through the university mobile application.
- Time-slot-based advance reservation tied to user academic schedules and registered vehicle identifiers.
- Duplication-prevention logic to ensure a single user cannot hold concurrent reservations.
- Automated security notification workflow for reservation violations.
- Advisory mechanism for redirecting external parking overflow based on detected idle reservations, as a partial mitigation of the informal sidewalk parking problem.

Out of Scope

- Physical infrastructure modifications such as constructing new parking areas, installing physical barriers, or expanding the current basement capacity.
- Management of car, bicycle, or other vehicle categories; the system exclusively addresses motorcycle parking.
- Legal or municipal enforcement of sidewalk parking on external public roads, which falls under the jurisdiction of urban traffic authorities and is beyond the university's software mandate.
- Parking facilities located in buildings or campuses other than the main engineering faculty facility

- Financial or billing systems; the SCPMS does not process payments or manage fee structures.

2.3 Stakeholder Identification and System Purpose Analysis

A stakeholder analysis was conducted to identify all parties whose behavior influences, or who are influenced by, the SCPMS. The following table summarizes each group, their functional role within the system, and their primary interest in its operation.

Stakeholder	Role in the System	Primary Interest
Students (motorcyclists)	Primary end users	Quick access, real-time space availability, reservation
Campus Security Staff	Monitors & enforces rules	Orderly allocation, violation reporting, reduced manual workload
Parking Administration	System owner & manager	Efficient space utilization, compliance tracking
Campus IT Department	Infrastructure provider	Reliable integration with existing university app & systems
University Management	Institutional authority	Improved campus mobility, user satisfaction, sustainability
Non-motorized Students	Indirectly affected users	Reduced congestion in shared pathways and corridors

System Purpose

The fundamental purpose of the SCPMS is to eliminate the information asymmetry that currently exists between actual parking availability and what users are able to observe. By replacing unstructured, first-come-first-served occupation with a structured reservation model, the system aims to achieve three interrelated outcomes: maximize the effective use of the fixed physical capacity in Basement 3; reduce the time and effort students expend searching for or waiting on spaces; and decrease the operational burden on security personnel who currently serve as the sole source of real-time guidance.

A secondary purpose — partially addressed through the External Overflow Advisory Module — is to create a data-driven mechanism for prioritizing who has legitimate access to informal external parking at any given time, transforming what is currently an unmanaged overflow problem into a rule-governed, transparent process.

2.4 Operating Environment Characterization

Physical Environment

- **Primary operating location:** Basement 3 (Sótano 3) Faculty of Engineering, Universidad Distrital Francisco José de Caldas, Bogotá, Colombia.
- **Space characteristics:** An enclosed underground facility currently lacking physical cell demarcation for motorcycles, with access controlled by security checkpoint at the entry ramp.
- **Connectivity requirements:** Wi-Fi or cellular network access for mobile application interaction; sensor endpoints (if IoT hardware is deployed in future phases) would require embedded network nodes within the basement.

Technological Environment

- **Host platform:** The university's existing mobile application, which already manages vehicle plate registration and access control. The SCPMS is designed as a functional extension of this application rather than a standalone product.
- **Backend dependencies:** University identity management system (for authentication), academic schedule database (for time-slot validation), and campus security communication infrastructure.
- **IoT readiness:** The current analysis phase treats the system as software-first. Physical IoT sensors for automated occupancy detection are identified as a future-phase enhancement; in the initial deployment, occupancy status may rely on user check-in/check-out actions and periodic security validation.

Operational and Temporal Environment

- **Operating hours:** Aligned with university academic schedule, typically 06:00–22:00 on weekdays, with reduced activity on weekends and academic breaks.
- **Peak demand periods:** Morning arrival (06:30–08:30), midday class transitions (10:00–12:00), and early afternoon (13:00–15:00), corresponding to major schedule change windows.

3. Data Collection Plan and Execution

3.1 Methodology Design

To support the systems analysis of the Smart Campus Parking Management System (SCPMS) described in Section 1, a structured primary data collection strategy was implemented. The objective of this phase was to gather empirical evidence about the current operational behavior of the motorcycle parking facility located in Basement 3 (Sótano 3) of the main campus building.
The collected data serves two main purposes:

- To validate the existence and magnitude of the inefficiencies described in the system overview.
- To provide quantitative and qualitative evidence that supports the need for the proposed digital parking management solution.

Following the workshop requirements, multiple complementary methods were used to ensure a reliable and comprehensive understanding of the system:

1. **Direct Observation** of the parking area and user behavior.
2. **Structured Surveys** directed to students and professors who use the facility.
3. **Usage Monitoring** to analyze parking frequency and duration patterns.
4. **Environmental Correlation** to connect parking demand with academic schedules.
5. **Process Documentation** to formally describe the current parking workflow.

3.2 Direct Observation

Direct observation was conducted in Basement 3 of the main campus parking facility, the area currently designated for motorcycle parking. The observation focused on variables such as spatial arrangement, occupancy levels, user behavior, search time, and spatial utilization.

3.2.1 Observed System Conditions

Observed Issue	Description
Lack of motorcycle demarcation	The parking area uses markings originally designed for cars rather than motorcycles.
Irregular parking alignment	Motorcycles are parked without a standardized arrangement.
Spatial inefficiency	Large unused gaps appear between vehicles.
Congestion during peak periods	Incoming users circulate searching for available spaces.

3.3 Structured Surveys

A structured survey was conducted among 19 members of the university community (students and professors) who regularly use the facility.

3.3.1 Participant Profile

Role	Number of Responses
Students	17
Professors	2

3.3.2 Transportation Method

Transportation Method	Responses
Motorcycle	16
Car	2
Bicycle	1

3.4 Usage Monitoring

Monitoring was conducted by analyzing frequency of use and parking duration.

3.4.1 Parking Usage Frequency

Frequency	Responses
Daily	9
3–4 times per week	6
1–2 times per week	3
Rarely	1

3.4.2 Typical Parking Duration

Duration	Responses
2-4 hours	6
4-6 hours	10
More than 6 hours	2

3.5 Environmental Correlation

Parking demand was analyzed in relation to the academic schedule.

Time Period	Observed Demand
06:30 - 08:30	High arrival rate associated with morning classes
10:00 - 12:00	Moderate turnover during class transitions
13:00 - 15:00	Second peak demand period

3.6 Process Documentation

The current parking workflow was documented as follows:

1. User arrives at the campus parking entrance.
2. Security personnel allow access to the basement.
3. User enters Basement 3 and searches for an available space.
4. User visually scans the area until an empty space is found.
5. If no space is available, the user waits or searches for external parking.

3.6.1 Identified Workflow Inefficiencies

Process Stage	Identified Issue
Parking entry	No information regarding available capacity
Space search	Users must manually inspect the parking area
Parking arrangement	Lack of standardized spatial allocation
Occupancy management	No system for reserving or prioritizing spaces

3.7 Primary Data Presentation

3.7.1 Frequency of Parking Saturation

Response	Count
Frequently	9
Sometimes	4
Rarely	3
Always	1

3.7.2 Perceived Space Utilization Efficiency

Response	Count
No	15
Yes	2
Not sure	2

3.7.3 Acceptance of a Digital Reservation System

Response	Count
Yes	17
Maybe	1
No	1

3.8 Data Quality Assessment

Although the survey sample size (19 participants) represents a limited subset of the population, the combination of observation and monitoring provides a consistent representation of the operational inefficiencies and supports the need for the **Smart Campus Parking Management System**.

4. System Analysis

The analysis focuses on the gap between the physical capacity of Basement 3 and its actual operational efficiency, which is degraded by both human and structural factors.

4.1 Analysis of the Current Workflow

Currently, the process is linear and lacks real-time data feedback:

- **Reliance on visual inspection:** Users must physically enter the basement to determine if there are vacancies, leading to unnecessary circulation and congestion at the access ramp.
- **Operational burden on security:** Security personnel act as the sole access filter and source of guidance, lacking tools that allow them to know exact availability without a physical walkthrough.
- **Reactive decision-making:** In the absence of available spots, users seek external alternatives (sidewalks) in a disorganized manner, shifting the mobility problem to public spaces.

4.2 Root Cause Identification (Inefficiency)

After analyzing the collected information, three main causes for the current system's failure have been determined:

1. **Information Asymmetry:** There is a total disconnect between what the system (parking lot) has available and what the actor (student/professor) knows before arriving.
2. **Spatial Disorder:** Due to the lack of demarcated motorcycle cells (using car markings instead), area utilization is sub-optimal. A single motorcycle can inadvertently occupy the space of three, reducing the perceived capacity.
3. **Absence of Prioritization:** The "first-come, first-served" model does not consider academic schedules, allowing vehicles to remain longer than necessary or block those with classes during peak hours.

4.3 Digital Solution Requirements

To transform this scenario, the technical analysis indicates that the **SCPMS** must fulfill the following:

- **Visibility (Data Input):** The system must transform the physical state of each cell into a binary digital datum (Occupied/Free) visible within the app.
- **Regulation (Business Logic):** Implementation of time slots directly linked to the academic schedule database to ensure vehicle rotation.
- **Traceability:** Mandatory linking of the reservation with the vehicle's license plate and the user's identity to facilitate audits and infraction reporting.

5. System Evaluation

The evaluation of the proposed system is based on its ability to transform a poorly managed physical resource into an efficient and traceable digital service.

5.1 Operational Impact Evaluation

- **Physical Space Optimization:** By implementing digital cell demarcation, the system eliminates the phenomenon where one motorcycle occupies the space of several, allowing the real maximum capacity of Basement 3 to be reached.
- **Reduction in Search Times:** Access to real-time availability via the mobile application reduces the flow of vehicles entering the basement without a guaranteed spot.
- **Academic Priority Management:** The use of time slots ensures that the parking lot is used by those with active academic activities, preventing passive or indefinite occupancy.

5.2 Security and Control Evaluation

- **Surveillance Automation:** The system decreases the subjective burden on security personnel by providing automatic alerts for unregistered vehicles or exceeded stay durations.
- **User Traceability:** Integration with university credentials and vehicle license plates allows for strict control over service users, increasing the security of the vehicle inventory on campus.

5.3 Technological Feasibility Evaluation

- **Infrastructure Integration:** As an extension of the university's existing mobile app, implementation does not require developing a platform from scratch, leveraging already available schedule and registration databases.
- **IoT Scalability:** Although the initial phase focuses on software management, the design is prepared to integrate physical sensors in the future (hardware phase), enabling full automation of occupancy monitoring.

5.4 External Impact Mitigation

- **Overflow Control:** Through the external advisory module, the system offers a partial but technical solution to the problem of motorcycles parked on sidewalks, prioritizing the use of inactive internal spaces during off-peak hours.

6. Identified Problems

Following the direct observation phase and survey analysis, critical issues have been identified that justify the technological intervention of the **SCPMS**.

6.1 Inefficiency in Physical Space Usage

- **Lack of specific demarcation:** The use of markings designed for cars in an area intended for motorcycles leads to chaotic distribution.
- **Wasted capacity:** Unnecessary gaps are created between vehicles, causing a single motorcycle to occupy the equivalent space of three potential cells.

- **False perception of saturation:** 79% of users believe the space is not used efficiently, creating a sense of lack of availability even when there are poorly utilized empty areas.

6.2 Failures in Access and Circulation Processes

- **Critical information asymmetry:** Users lack prior data on availability, forcing them to enter the basement solely to verify if a spot exists.
- **Peak hour bottlenecks:** High arrival rates between 06:30 and 08:30 cause the access ramp to collapse due to manual spot searching.
- **Excessive search times:** Drivers must perform repetitive visual scans within the facility, increasing internal congestion.

6.3 Deficiencies in Management and Control

- **Manual operational burden:** Security personnel rely entirely on visual judgment and intermittent physical supervision to maintain order.
- **Indefinite occupancy:** There is no mechanism to limit stay duration, allowing vehicles to remain parked beyond their actual academic needs.
- **Uncontrolled overflow:** The lack of managed alternatives pushes users to park on sidewalks and public roads, creating an external mobility problem that the university currently cannot regulate.

6.4 Current System Vulnerabilities

- **Lack of traceability:** There is no digital record linking the vehicle's license plate with its exact location and the user's identity in real time.
- **Non-compliance with regulations:** 53% of survey respondents state that the parking lot becomes saturated "Frequently" or "Always," which encourages the disregard of internal circulation zones due to user frustration.

7. Improvement Proposals

The proposed improvements are divided into three strategic axes designed to tackle disorganization, lack of information, and the current operational burden.

7.1 Digitalization and Space Structuring

- **Virtual Cell Demarcation:** Replace the use of car markings with a digital blueprint featuring specifically numbered spaces for motorcycles, eliminating the waste of physical area.
- **Real-Time Visualization:** Implement an interactive map within the mobile application that displays the binary status (vacant/occupied) of each cell, eradicating the need for physical visual inspection.

7.2 Time Management Optimization

- **Schedules-Based Reservations:** Link parking requests to the university's class database to ensure that space is prioritized for those with active academic activities.
- **Time-Slot Occupancy Control:** Establish specific time windows (e.g., 06:00–10:00) to prevent indefinite occupancy and encourage vehicle rotation during peak demand periods.
- **Duplicity Prevention:** Apply business logic that prevents a single user from holding multiple active reservations, ensuring equitable access to the resource.

7.3 Strengthening Control and Security

- **Automated Infraction Notifications:** Create an instant alert flow for security personnel whenever an unregistered vehicle occupies a cell or exceeds its reservation time.
- **Plate and Credential Identification:** Integrate the system with existing vehicle data in the university app to ensure full traceability for every occupant.

7.4 External Overflow Management

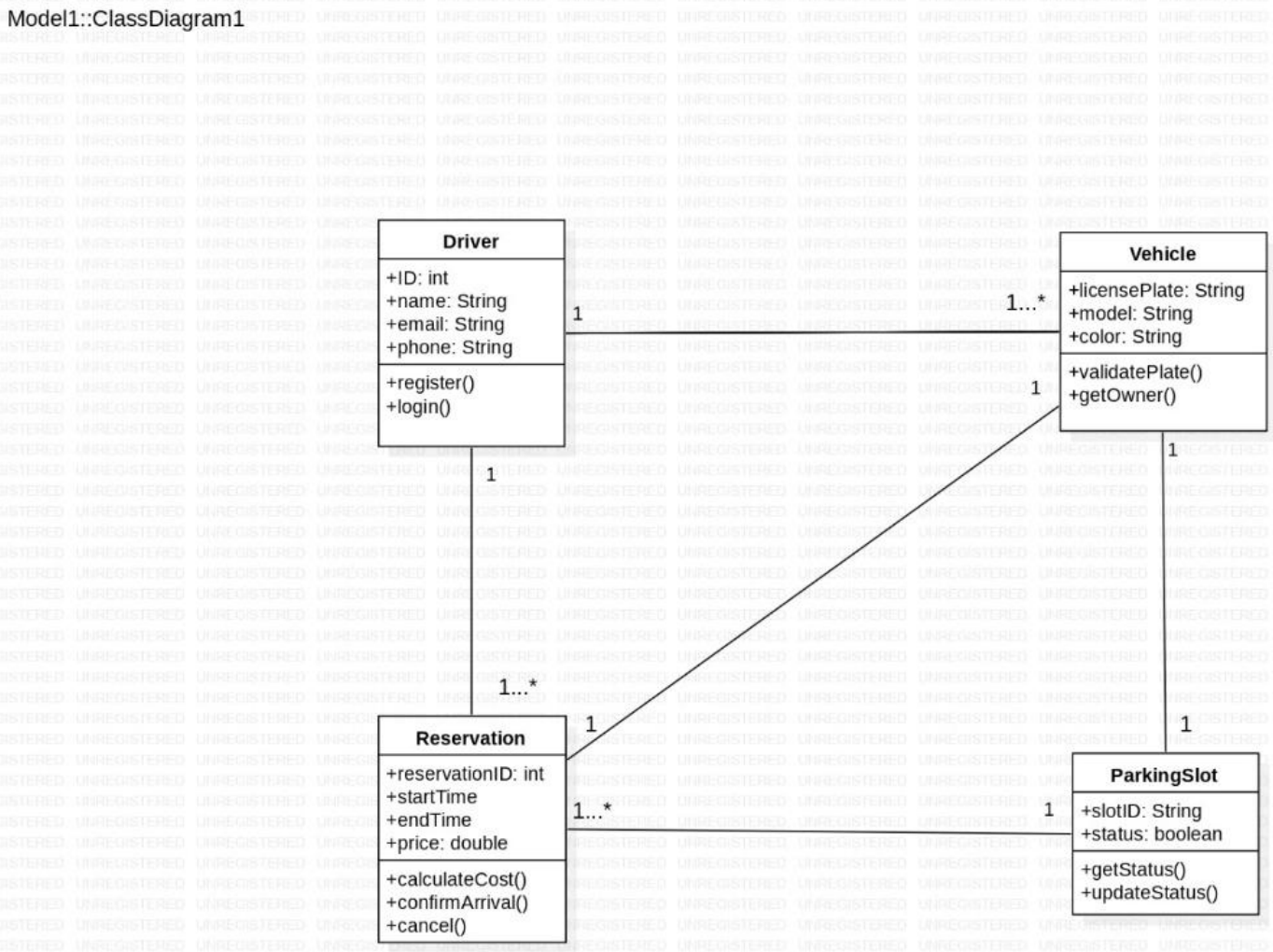
- **Informal Parking Advisory Module:** Utilize inactive reservation data to inform users of actual availability, technically and orderly reducing disorganized parking on surrounding sidewalks.

8. Conclusions

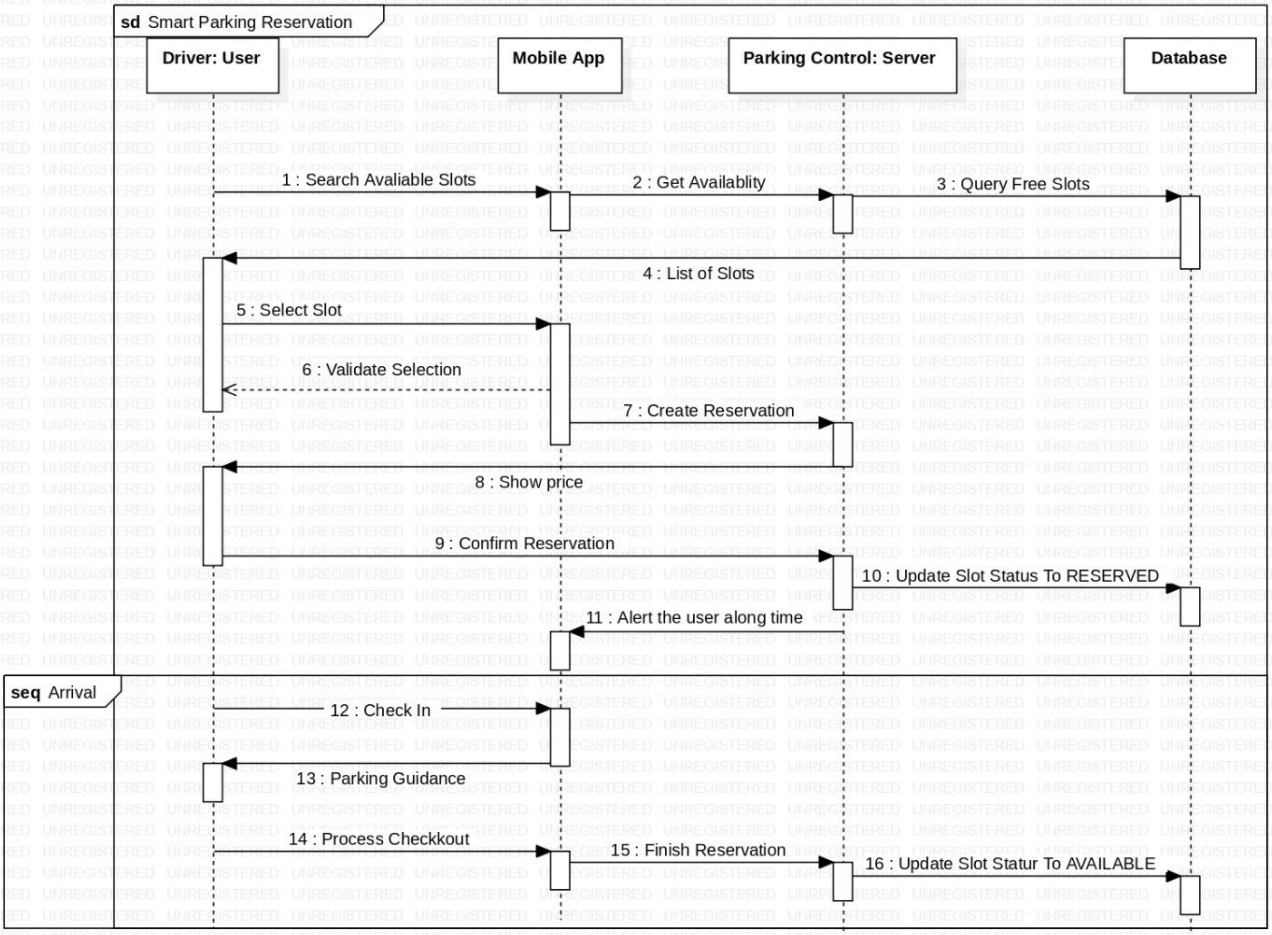
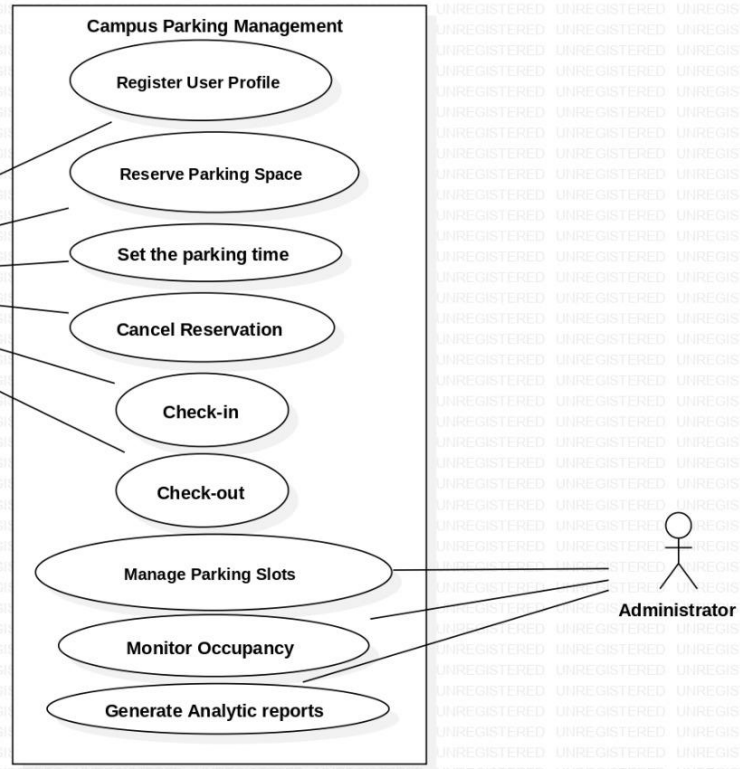
The development of this analysis allows for the extraction of the following fundamental conclusions regarding the feasibility and success of the project:

- **Validation of Technological Need:** Field research confirmed that 79% of users perceive a lack of efficiency in the current use of space, validating the urgency of moving from an improvised occupancy model to one based on digital management.
- **Optimization without Physical Infrastructure:** It was determined that the issue in Basement 3 is not a lack of total area, but rather spatial disorder where one motorcycle can occupy the space of three. The **SCPMS** resolves this through virtual demarcation and business logic without incurring construction costs.
- **Elimination of Information Asymmetry:** The system fulfills its fundamental purpose by providing real-time visibility of availability. This drastically reduces search times and congestion during the identified peak demand periods (especially between 06:30 and 08:30).
- **Sustainability and Equity:** By linking reservations with academic schedules and applying duplicity prevention logic, fair access to the resource is guaranteed for the entire university community, maximizing daily vehicle rotation.
- **Scalability and Security:** Integration with the existing mobile application and the automated alert system not only optimizes the work of security personnel but also establishes the technological foundation for future IoT sensor implementation.

9. Visual representation of the system (diagrams)



Collaboration2::Smart Parking Reservation::Parking Reservation



Model1::UseCaseDiagram1